



# A methodology for high resolution vehicular emissions inventories in metropolitan areas: Evaluating the effect of automotive technologies improvement

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## ABSTRACT

The establishment of public policies that successfully control pollutants concentration in the atmosphere has been a challenge in many urban areas. As the road transport is an important source of pollution, information on fleet composition and its emissions are important to verify emissions control strategies and to guide mitigation measures. In that context, the present study proposes a method to develop and evaluate high resolution inventories for road transport emissions in metropolitan areas. This method was applied to the Florianópolis Metropolitan Area (FLMA), in Santa Catarina, Brazil. A probabilistic bottom-up approach was employed using modeled transportation data to produce a vehicular emissions inventory in high resolution. This approach was developed to account for fleet characteristics in emission factors estimations. The impact of vehicle categories and their emissions distribution was also analyzed. Results showed that light vehicles categories were responsible for 96% of CO, 87% of NMHC, 82% of CH<sub>4</sub> and 74% of N<sub>2</sub>O emissions, well distributed throughout the road network. Heavy-duty vehicles emitted the biggest amount of the NO<sub>x</sub> and MP (80% and 94%, respectively), mostly concentrated in expressways. An analysis of emissions on each network hierarchy showed that expressways comprehended 45% of CO, 48% of HC, 79% of NO<sub>x</sub>, and 85% of PM emissions. The residual part of those pollutants is emitted by local, collector, connector and arterial roads. And finally, a hypothetical scenario pointed that replacing the FLMA's fleet for vehicles of model-year from 2017 has a potential to decrease emissions up to 95%.

## 1. Introduction

In air quality management systems, usually the first step to control vehicular pollution is to elaborate an emissions inventory (Ferreira et al., 2013; Trombetti et al., 2018). Inventories guide air quality intervention measures, quantify the main substances emitted, identify the best available control technologies and provide future perspectives (Borge et al., 2008; Ferreira et al., 2013; Gulia et al., 2015; Iodice and Senatore, 2015). Literature recommends that vehicular emissions inventories should include pollutants' spatial distribution and temporal variability. However, in many developing countries, mobile sources emissions inventories are scarce and only account for total emissions, mainly due to the absence of a method to deal with data scarcity (Borge et al., 2014; Ferreira

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et al., 2013; Saide et al., 2009; Tuia et al., 2007). There is still room for improvement of emissions inventories in metropolitan areas (Andrade et al., 2017). The present work proposes a method to develop and evaluate high resolution inventories (spatial and temporal) for road transport emissions from metropolitan areas. Through this method, it is possible to find the major contributors of each pollutant, most pollutant roads and regions. In order to estimate emissions, a combination of a probabilistic bottom-up approach and vehicular flow data from the modeling system *TransCAD* is employed (Caliper, 2019). Also, the effect of improvement of automotive technologies on reducing atmospheric emissions is tested through a hypothetical scenario. This work evaluates emissions abatement due to fleet technology changing on the level of the roads and areas ( $1 \text{ km}^2$ ).

In the context of public health, it is well known that ambient air pollution has been associated to several adverse effects on human health (EMEP/EEA, 2013; Tam et al., 2015; Torres et al., 2018). The United Nations Environmental Program estimates that developed countries plus India and China spend US\$3.5 trillion per year in public health and death related costs associated to air pollution (UNEP, 2014). In urban centers, vehicular emissions are responsible for a significant part of pollutants found in the local atmosphere (Ferreira et al., 2013; Lumbreras et al., 2014; Mage et al., 1996; Shahbazi et al., 2016). The establishment and effectiveness of regulations towards vehicular emissions reduction are primordial to improve environmental health. Therefore, effective emissions control strategies should be based on a solid air quality management system, justifying the development of robust emission inventories as proposed in the present work.

According to Andrade et al. (2017), many successful programs implemented by governmental agencies have reduced mobile sources emissions. In the metropolitan area of the megacity São Paulo, public policies for emissions reduction have shown a positive effect on reducing primary pollutants such as CO, NO<sub>x</sub> and PM<sub>10</sub>. However, the ozone concentrations have not followed the same trend and the formation of secondary particles is still a concern in the Metropolitan Area of São Paulo (Andrade et al., 2017; Salvo et al., 2017). The issues associating air quality to urban mobility in metropolitan areas need to be further studied.

In this work, the proposed method is applied in the Florianópolis Metropolitan Area (FLMA), to elaborate the first vehicular emissions inventory in high spatial resolution in the State of Santa Catarina, Brazil. Florianópolis presents the 3rd highest human development index among cities in Brazil (IBGE, 2017), and is recognized as the most innovative capital in the country (Ewers et al., 2015). Urban mobility in Florianópolis is still an important concern to be addressed. From the resulting inventory, this study analyzes FLMA's fleet composition, and its alignment to the current Brazilian regulations.

## 2. Materials and methods

This section details the method proposed for high resolution vehicular emissions inventories elaboration and evaluation, applying it to the FLMA. In order to elaborate the inventory, data on vehicular flow were collected from a detailed transportation model prepared as part of the Sustainable Urban Mobility Plan of the Great Florianópolis Area (PLAMUS) (PLAMUS, 2015). Data on fleet composition were also collected, to produce an emissions inventory based on vehicles category, subcategory, model-year and fuel. A probabilistic bottom-up method was employed to calculate the weighted emission factor of each category. To aggregate a spatial resolution to the inventory, the method was applied street-by-street within the geographically referenced road network provided by PLAMUS. Finally, to analyze FLMA's fleet composition and its alignment to the Brazilian policies for emissions control, this section presents a brief description of how the implementation of PROCONVE's phases was evaluated.

### 2.1. Study area

The study area was called the Florianópolis Metropolitan Area (FLMA), which includes five of the main cities of the official Florianópolis Metropolitan Region, within the mesoregion of the Great Florianópolis. The geographical center of FLMA is located at coordinates 27.59 S and 48.54 W. The Santa Catarina State's legislation includes four other cities to the Florianópolis Metropolitan Region, which were not considered within FLMA. The included cities are Biguaçu, Governador Celso Ramos, Palhoça, São José and the capital of the State of Santa Catarina, Florianópolis (Table 1).

According to the estimates of the Brazilian Institute of Geography and Statistics (IBGE) for 2017, these cities together have 971,269 inhabitants, a territory of  $1,706 \text{ km}^2$  and 665,262 licensed vehicles (IBGE, 2017). Florianópolis has half of the population and the fleet of FLMA, and the largest territory within the study area. Florianópolis also have the highest fleet per capita of the group (0.71), followed by São José and Palhoça (0.67), Biguaçu (0.65) and finally Governador Celso Ramos (0.55). Moreover, Florianópolis's population is 96% concentrated in urban area and has the third largest fleet per capita in Brazil (IBGE, 2010), therefore a huge part of its population lives and transits near pollution sources. São José is the municipality that presents the highest population density ( $1593 \text{ inhabitants/km}^2$ ), twice as much as Florianópolis. Thus, the cities within the study area have similar rate of vehicles

**Table 1**  
Territory ( $\text{km}^2$ ), population (inhabitants) and vehicular fleet (road vehicles) of the municipalities within the FLMA.

| Municipality           | Territory | Population | Fleet   |
|------------------------|-----------|------------|---------|
| Florianópolis          | 675       | 485,838    | 343,392 |
| São José               | 150       | 239,718    | 160,236 |
| Palhoça                | 395       | 164,926    | 110,281 |
| Biguaçu                | 368       | 66,558     | 43,560  |
| Governador Celso Ramos | 117       | 14,229     | 7793    |

per capita, except for the city of Governador Celso Ramos, which has a smaller fleet and population density.

In Brazil, vehicular emissions reduction is regulated by the Brazilian National Council for the Environment (CONAMA). The CONAMA regulates emissions from light-duty, commercial light and heavy-duty vehicles through a program established in 1986 known as PROCONVE. Similarly, CONAMA has established in 2002 a program called PROMOT to regulate motorcycles and similar vehicles. Both programs are continuously updated, demanding that new vehicles perform better than old vehicles. Every update is a phase with a defined implementation period and goals, which becomes increasingly restrictive in terms of pollutant emissions. The air contaminants restricted are Carbon Monoxide (CO), Total Hydrocarbons (HC), Non-methane Hydrocarbons (NMHC), Methane (CH<sub>4</sub>), Aldehydes (RCHO), Nitrogen Oxides (NO<sub>x</sub>) and Particulate Matter (PM). For light-duty and commercial-light categories, implementation phases range from L1 to L6; for heavy-duty vehicles, from P2 to P7; and for motorcycles, from M1 to M4 (CONAMA, 1986, 2002). Through PROCONVE and PROMOT, public policies have directly influenced the improvement of automotive technologies and fuel efficiency in Brazil.

Phase L1 was implemented in 1988, establishing the first restriction limits for CO, HC and NO<sub>x</sub>, which has influenced the implementation of exhaust gases recycling, secondary air injection and ignition timing optimization. In 1992, phase L2 started and forced adapting catalytic converters and electronic fuel injection for ethanol addition. At that point, Brazilian standards were far less restrictive than the European Union regulations (Euro series). In 1994, emission rates started to be restricted for heavy-duty vehicles in phase P2. However, P2 did not contribute to reduce emission levels, as limit values were not restrictive enough. On the same year, P2 was replaced by phase P3, with stricter limits for CO, HC and NO<sub>x</sub>, which resulted in increase of vehicles autonomy and horsepower. Only in 1997, with phase L3, PM emissions started to be restricted, and the commercial-light category had to implement the same standards than light-duty vehicles. L3 limits were based on Euro 2 standards. Stricter limits forced manufacturers to implement some of the best technologies available, such as the lambda sensor. In 1998, phase P4 was established to increase restrictions on PM emissions. In 2003, with phase P5, restrictions increased even more as they were based on Euro III. Only in 2003, restrictions on motorcycles emissions started. From phase M1 to M4, limits established for CO have decreased 85%; for HC up to 92%; and around 50% for NO<sub>x</sub>. From 2005 to 2012, phases L4 and L5 focused in reducing HC and NO<sub>x</sub> emissions based on Euro 3/4 standards. Starting from 2013, phase L6 was based in Euro 5 - except for PM limits, and encouraged the introduction of catalytic converters by oxidation and improvement of gases recycle. P6 was the only phase that could not be implemented, being replaced by phase P7 in 2012, based on Euro V regulations. In November of 2018, the next phases of PROCONVE - L7 and P8 - have been established for initiating in 2022, and L8, in 2025 (Brazil, 2013; CETESB, 2014; Dieselnets, 2019).

As established by CONAMA, every state is responsible for elaborating its own Vehicular Pollution Control Plan (VPCP). However, the State of Santa Catarina (SC) has not complied with creating its VPCP yet. The Brazilian Federal Justice has already condemned SC to develop its plan, as any provision to elaborate the VPCP has not been taken since 2011 (MPF, 2017). The air quality diagnosis of SC being unknown can represent a risk to human health and well-being in the state.

## 2.2. Vehicular flow and transport network

The transport network, along with the flow of vehicles, were obtained from a detailed transport model developed for the Sustainable Urban Mobility Plan of the Great Florianópolis Area (PLAMUS, 2015). The plan was the result of a rigorous study performed in the Great Florianópolis Area by the Government of the State of Santa Catarina, with the participation of the numerous institutions, such as universities, municipalities, consultants, public transport operators, and others. The model of the transport network was developed in *TransCAD*, a Geographical Information System (GIS) tailored for transport applications. Traffic was modeled for the peak hours in the morning and evenings. The plan categorized vehicles in: private, light trucks and heavy trucks. Detailed information on the method and modeling conditions employed by PLAMUS are available at: <https://observatoriadamobilidadeurbana.ufsc.br/publicacoes/>.

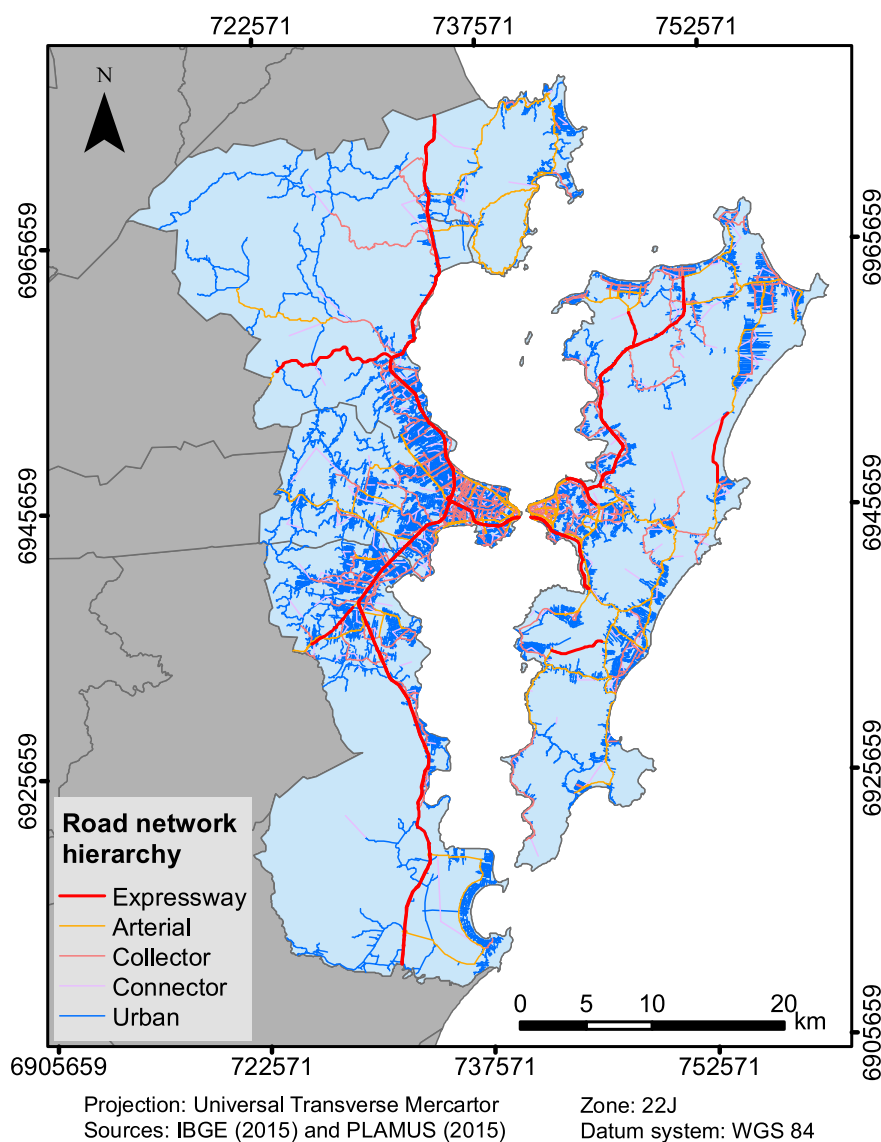
The transport network was categorized in expressways, connectors, arterials, collectors, and urban roads, as shown in Fig. 1. Expressways included federal and state roads, designed for high speed, intense flow of vehicles and uninterrupted travel, which may be connected by roads called “connectors”. Just like expressways, arterials were considered to carry large volumes of traffic at high speed, except that they have occasional intersections and traffic lights. Collectors included roads that collect traffic from arterials and distribute to urban roads. Urban roads included streets that carry low volume of traffic under low speed limits.

## 2.3. Fleet data

The traffic flow in *TransCAD* was not segregated by category and subcategory, neither by model-year or fuel utilization according to emission factor (EF) categories. Instead, it was segregated in private vehicles and light and heavy trucks. Therefore, licensing and purchasing data were used to disaggregate vehicular flow information, providing the percentage of vehicles in each category. To organize vehicular flow data according to EFs from CETESB, private vehicles were split in light-duty, commercial-light and motorcycles. Light and heavy trucks were combined into heavy-duty vehicles.

The categorized amount of vehicles was obtained from the National Registry of Automobile Vehicles (RENAVAM), elaborated by the National Department of Traffic (DENATRAN, 2017). Categories from RENAVAM were arranged in light-duty, commercial light, motorcycles and heavy-duty to correspond to EF categories from CETESB. Emission factors also depend on the vehicle subcategory. Although light-duty and commercial-light categories are not segregated in subcategories, motorcycles and heavy-duty vehicles are.

Motorcycles EFs are subcategorized by cylinder capacity (cc). Until 2009, motorcycles subcategories were: (a) equal or below to 150 cc, (b) from 151 to 500 cc, and (c) above 501 cc. From 2010 onwards, subcategories are: (a) equal or below to 150 cc, and (b)



**Fig. 1.** Road network from the Florianópolis Metropolitan Area (FLMA), . adapted from [PLAMUS \(2015\)](#)

above 150 cc. In order to estimate motorcycles in each subcategory, purchasing and licensing data were obtained from the Brazilian Association for Motorcycles and Similar Manufacturers ([ABRACICLO, 2016](#)).

Heavy-duty vehicles are subcategorized according to their total gross weight. For trucks, subcategories are: (a) demi-light, (b) light, (c) average, (d) demi-heavy, and (e) heavy, depending on their total gross weight. The amount of trucks and their subcategory definitions were obtained from the National Association for Automobile Vehicles ([ANFAVEA, 2017](#)). This method was not applied to buses, due to unavailable information about buses routes. Since the PLAMUS elaboration in 2015, another company is responsible for operating the public transport of Florianópolis. Therefore, new routes have been implemented and old ones were altered. A few route alterations were made to the other cities within the FLMA as well. The route system used by PLAMUS (from 2013) is now outdated. In that context, the present study does not include emissions from buses.

Details on model-year for each category were obtained from RENAAM (DENATRAN, 2017). For light-duty vehicles, EFs from CETESB range from 1982 to 2018. For commercial-light, EFs vary from 1983 to 2018. For motorcycles, from 2003 to 2018. And for heavy-duty vehicles, from 1999 to 2018.

Fuel utilization data were also from RENAAM (DENATRAN, 2017). Ethanol, gasoline, or diesel were considered as fuels. EF from CETESB include, for light-duty vehicles from 1982 to 2002: (a) gasoline or (b) ethanol. From 2003 to 2006 the options (c) flexible-fuel using gasoline; and (d) flexible-fuel using ethanol were included. Starting from 2007 the option (b) ethanol was suppressed. For commercial-light vehicles, from 1983 to 2002 the options were: (a) gasoline or (b) ethanol. From 2003 to 2005 the options (c)

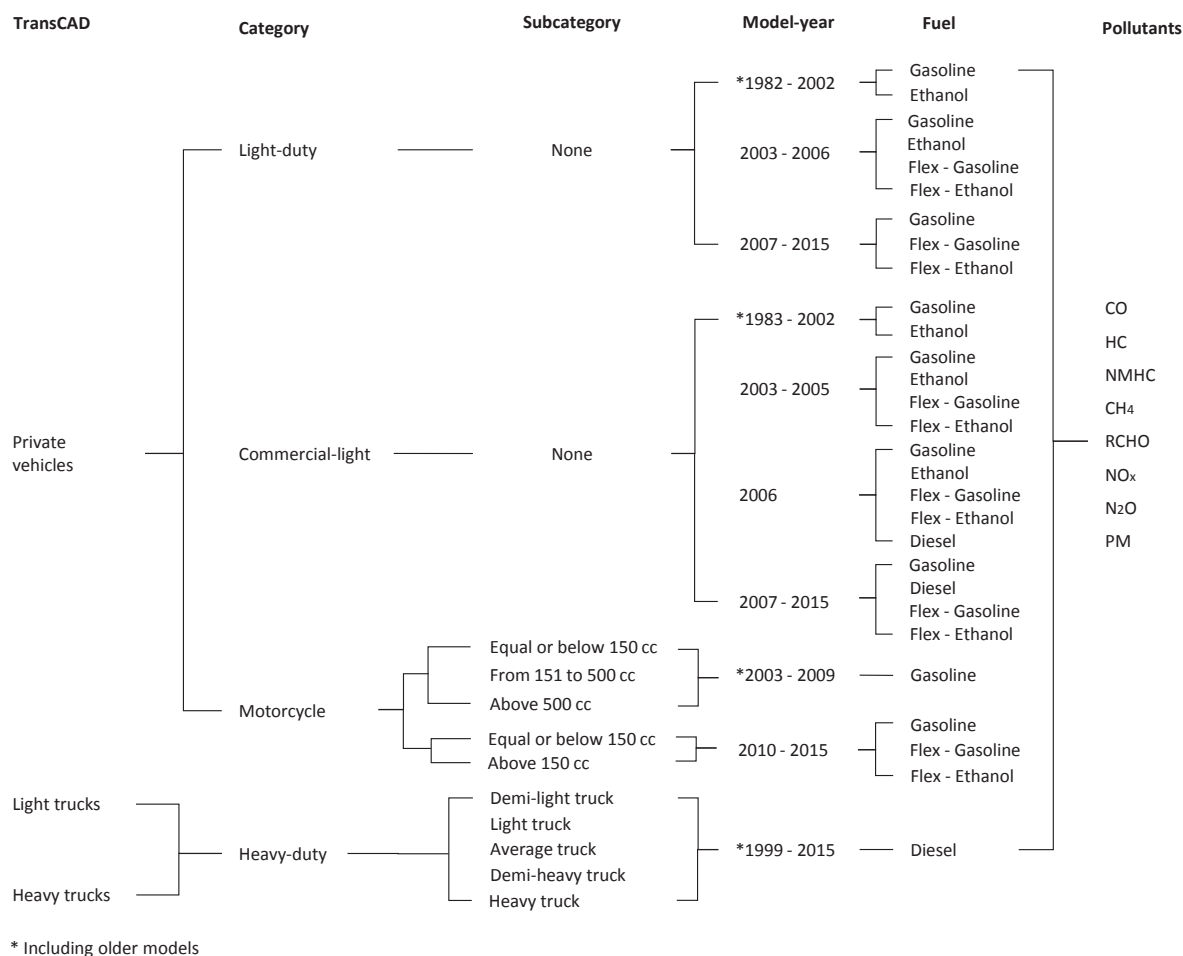


Fig. 2. Summary of vehicles data on categories, subcategories, fuel type and model-year.

flexible-fuel using gasoline and (d) flexible-fuel using ethanol were included. In 2006, the option (e) diesel was also added. From 2007 onwards, the option (b) ethanol was suppressed. For motorcycles, from 2003 to 2009 the option was gasoline. Starting from 2010, (c) flexible-fuel using gasoline and (d) flexible-fuel using ethanol were included. For heavy-duty vehicles, diesel is the unique fuel option considered.

Vehicles fueled with natural gas were not accounted for, due to their small representation and low emission potential. Concerning flexible-fuel vehicles, a segregation factor was applied. This segregation factor is subjected to change as a function of fuel price regulations in Brazil, therefore its adoption was based on an analysis of sales price and fuel consumption (Goldemberg et al., 2008). The curve presented by Goldemberg et al. (2008) was adjusted to the average sales prices of ethanol and gasoline in the State of SC in 2017, provided by the Brazilian National Agency for Oil, Natural Gas and Biofuel (ANP). It was considered that in 2017, 2% of flexible-fuel vehicles in FLMA used ethanol; and 98%, gasoline.

The diagram in Fig. 2 summarizes fleet data on categories, subcategories, fuel type and model-year, as well as the pollutants considered in this study. The number of vehicles for each category, and according to fuel type and model-year are detailed in Appendix A.

## 2.4. Probabilistic bottom-up method

### 2.4.1. Emissions estimation method

The classical bottom-up equation has been widely used for estimating emissions in regional and local scales. However, some adaptations are required to estimate emissions per segment of the road, which is ideal for high resolution inventories (Cook et al., 2006; Gómez et al., 2018; Policarpo et al., 2018; Singh et al., 2018). The classical bottom-up method is represented by the following formula, adapted from (Cook et al., 2008):

$$E_p = EF_p * N * D \quad (1)$$

with  $E_p$  (mass per unit of time) the emission rate of pollutant  $p$ ,  $EF_p$  (mass per distance traveled) the emission factor from vehicular

sources for pollutant  $p$ ,  $N$  (number of vehicles per unit of time) the flow of vehicles, and  $D$  (units of length) the distance traveled in the road-link.

Formula (1) was applied to calculate exhaust emissions on each road link of the FLMA, with modeled vehicular flow and the FLMA's fleet characteristics. The flow of vehicles  $N$  on each road with distance  $D$  was determined by a vehicular flow model detailed in Section 2.2. In relation to emission factors a probabilistic method detailed in Section 2.4.2 was developed to compute the average values for each pollutant, according to fleet characteristics detailed in Section 2.3. The cities in the FLMA were considered to have similar fleet composition, where the city of Florianópolis was used as reference.

The present method can be applied to estimate exhaust emissions. Vehicular speed influences, evaporative losses caused by fuel storage and refueling, crankcase gases, brakes and tyres wear, and soil resuspension emissions were not estimated. Even though these sources of emissions may be significant, they are not on the scope of this work and may be included in future works.

#### 2.4.2. Weighted emission factors according to fleet composition

The EFs were obtained from samplings conducted by the Environmental Company of São Paulo State (CETESB, 2019). CETESB provides emission factors of new vehicles for the categories light-duty, commercial light, motorcycles and heavy-duty vehicles, as presented in Appendix B. A probabilistic method described next was developed to produce a weighted EF for each category and subcategory. The developed method accounts for the fleet composition in terms of category, subcategory, model-year and fuel utilization. It is important to mention that this method can be applied either to EFs sampled for new vehicles, or EFs that include deterioration factors.

As vehicular inspection for emissions control is not conducted in the FLMA, EF did not account for vehicles deterioration according to the age of the vehicle. Instead, emission factors developed for new vehicles were applied for the FLMA's fleet, according to their Model Year. If the user opt to use deteriorated emission factors, it has to be done prior to the EF weighting.

For light-duty vehicles, the weighted EF for each pollutant  $p$  was calculated by:

$$\bar{EF}_{LD,p} = \sum_{y=1}^n \sum_{f=1}^n EF_{p,y,f} \cdot P(y|f) \quad (2)$$

with  $P(y|f)$  the conditional probability of a vehicle to be of model-year  $y$ , using fuel  $f$ , and  $EF_{p,y,f}$  the EF of year  $y$  and fuel  $f$ , for pollutant  $p$ . Eq. (2) was also applied to commercial light category.

For motorcycles, the EF is also dependent on cylinder capacity, thus:

$$\bar{EF}_{M,p} = \sum_{y=1}^n \sum_{cc=1}^n \sum_{f=1}^n EF_{p,y,cc,f} \cdot P(y|f) \cdot P(y|cc) \quad (3)$$

with  $P(y|f)$  the conditional probability of a motorcycle to be of model-year  $y$ , using fuel  $f$ ;  $P(y|cc)$  is the probability of a motorcycle to be of model-year  $y$  and be of  $cc$  cylinder capacity, and  $EF_{p,y,cc,f}$  is the EF of year  $y$ , cylinder capacity  $c$ , and fuel  $f$ , for pollutant  $p$ .

Heavy-duty vehicles are also dependent on subcategory, so the average EF for each pollutant  $p$  was calculated by:

$$\bar{EF}_{HD,p} = \sum_{y=1}^n \sum_{s=1}^n EF_{p,y,s} \cdot P(y|s) \quad (4)$$

with  $P(y|s)$  the conditional probability of a heavy-duty vehicle to be of model-year  $y$ , and be of subcategory  $s$ , for pollutant  $p$ .

Weighted EFs can be useful parameters for vehicular pollution control, guiding intervention measures. Ibarra-Espinosa et al. (2018) developed VEIN 0.2.2, an open source model that provides hourly vehicular emissions estimation at road-link level. VEIN 0.2.2 calculates vehicular flow at street level for each fraction of vehicle type, model-year and fuel, then exhaust emissions are calculated for each fraction of vehicular flow (Ibarra-espinosa et al., 2018). The main difference from the method presented by this paper and the cited work is the weighted EFs calculation, which reflects vehicular fleet composition. Weighted EFs are intermediate results that provide a general idea of how close each vehicular category is from meeting national standards. Also, VEIN 0.2.2 is a more robust model, which requires refined input data, often not available in regions where data is scarce.

#### 2.5. Spatial distribution

Emissions were distributed throughout the network provided by PLAMUS using transport data obtained from a TransCAD transportation model of the FLMA. The transportation model contains vehicular flow data and road-link length for each road in a vector file. For each road referenced on the *shapefile*, formula (1) was applied with average emission factors calculated as described in Section 2.4.2, according to fleet characteristics detailed in Section 2.3.

#### 2.6. Policies implementation strategy and scenarios

According to the Brazilian National Road Vehicle Inventory of 2013, the continuity of PROCONVE should be associated to their effectiveness in improving ambient air quality (Brazil, 2013). Road transport emissions are often associated to air quality deterioration in urban centers, therefore the authors have chosen to evaluate PROCONVE effectiveness by analyzing its impact on



**Table 2**Weighted EFs ( $\text{g km}^{-1}$ ) of the FLMA for each pollutant, according to vehicle category.

| Category         | CO    | HC    | NMHC  | CH <sub>4</sub> | RCHO  | NO <sub>x</sub> | N <sub>2</sub> O | PM    |
|------------------|-------|-------|-------|-----------------|-------|-----------------|------------------|-------|
| Light-duty       | 2.927 | 0.293 | 0.243 | 0.049           | 0.008 | 0.207           | 0.019            | 0.001 |
| Commercial light | 3.391 | 0.323 | 0.267 | 0.055           | 0.008 | 0.244           | 0.021            | 0.003 |
| Motorcycles      | 1.825 | 0.284 | 0.238 | 0.046           | –     | 0.075           | 0.001            | 0.004 |
| Heavy-duty       | 0.700 | 0.255 | 0.195 | 0.060           | –     | 4.146           | 0.030            | 0.154 |

reducing vehicular emissions.

As shown in formula (1), vehicles emissions potential is represented by their weighted emission factor. Therefore, analyzing the evolution of EF values over the years is a way of verifying the influence of policies implementation in vehicular and fuel technologies. The influence of Brazilian regulations was approached by comparing the FLMA's current fleet weighted average EFs to those obtained by CETESB.

The lowest emission would be reached if the FLMA's weighted EF was equal to emission factors measured for vehicles of 2017's model-year (most recent sampled EF at that time). Therefore, this paper also assessed the potential of emissions reduction by updating the FLMA's fleet to the newest technology available, applying EFs from 2017 to a hypothetical scenario.

### 3. Results and discussion

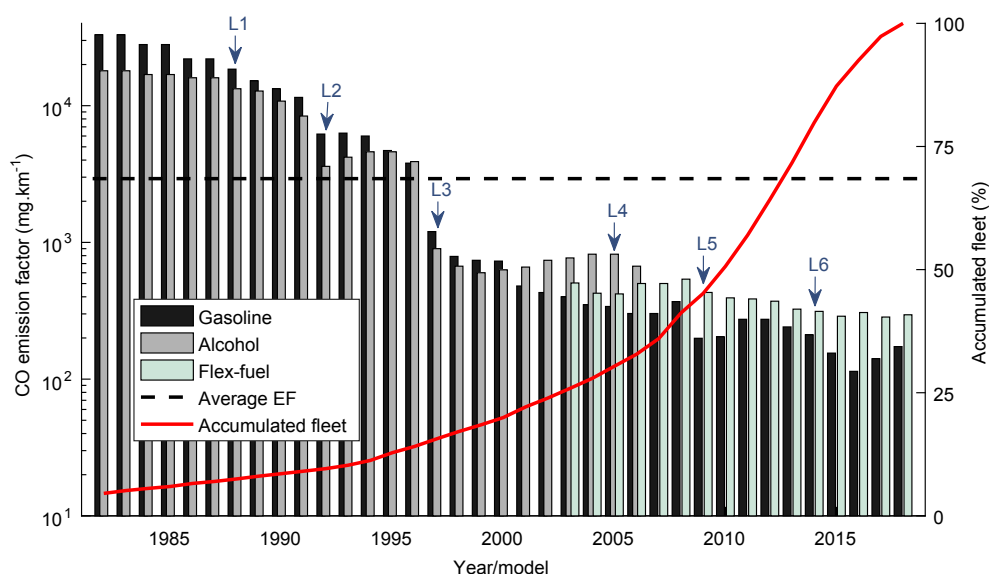
This section presents the results of the proposed method in the FLMA. First the results on weighted emission factors and their alignment to PROCONVE's phases are presented. Then, emissions from each category are shown and the total compared to inventories from different regions. Next, the total emissions distributed throughout the road network are presented, enabling the analysis of emissions according to road hierarchy. Finally, an analysis of emissions reduction potential of the hypothetical scenario is proposed in Section 2.6.

#### 3.1. Weighted emission factors for the FLMA

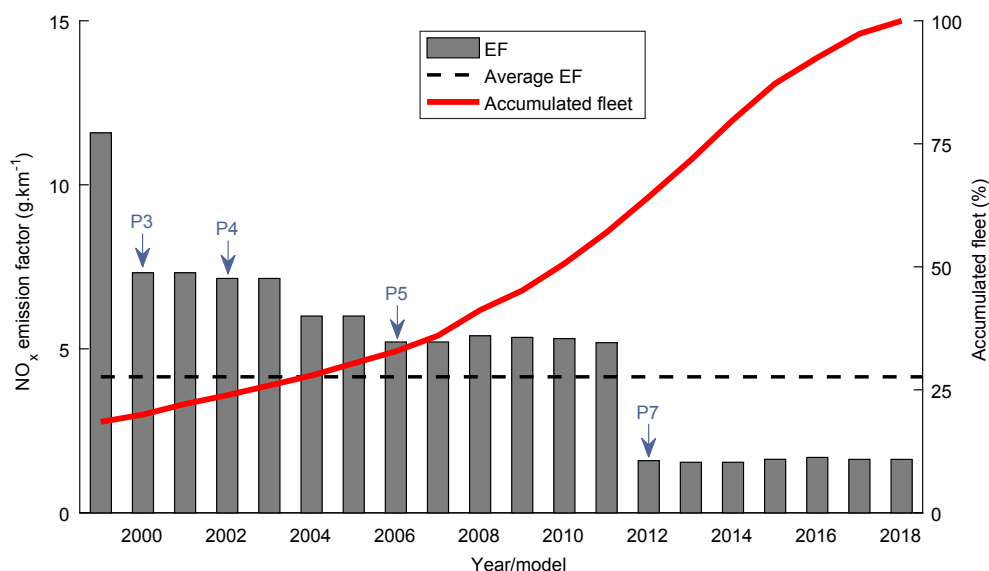
The weighted EFs from the FLMA's fleet were calculated by applying formulas (2) and (4) to the data on emission factors and fleet, as described in Sections 2.4.2. The results are presented in Table 2 for each vehicle category and by pollutant.

Figs. 3 and 4 illustrate the effectiveness of Brazilian policies implementation by comparing the EF from annual samplings described in Section 2.4.2 and the weighted EFs for the FLMA's fleet shown in Table 2.

Fig. 3 presents CO EFs sampled from light-duty vehicles, from 1982 to 2018. EFs are represented by bars, for vehicles that use either alcohol, or gasoline or that are flexible-fuel vehicles. The average CO EF of the FLMA's light-duty vehicles is represented by a dashed line along the entire period of the implementation of PROCONVE. PROCONVE's phases are represented by text arrows, indicating the initial year of each phase implementation. The red curve shows the cumulative percentage of current light-duty



**Fig. 3.** Carbon monoxide EF evolution per type of fuel for light-duty vehicles (left axis) and accumulated percentage of licensed vehicles in the FLMA, in relation to FLMA's 2017 fleet (right axis).



**Fig. 4.** Nitrogen oxides EF evolution for heavy-duty vehicles (left axis) and accumulated percentage of licensed vehicles in the FLMA, in relation to FLMA's 2017 fleet (right axis).

vehicles in the FLMA, licensed until the corresponding year. As indicated by the bars in the figure, EFs sampled prior to L1 implementation did not decrease significantly. EFs showed significant decrease from L1 to L3. Phases L4, L5 and L6 did not express effectiveness on EF reduction. As indicated by the red curve, over 50% of Florianópolis fleet has model-year from 2010. However, the dashed line in Fig. 3 shows that the weighted average EF for CO corresponds to the emission factor of a model-year from 1996, by the end of phase L2. Ideally, the dashed line should be closer to the EF represented by the bars in L6 instead. This suggests that, even though the light-duty fleet from the FLMA is relatively recent, older vehicles have a significant impact on the FLMA's average EF for CO.

Besides CO emission factors, other pollutants' EFs were assessed for light vehicles. For HC, the average EF for the light-duty category is equivalent to a vehicle from around 1997 (L3); and for NO<sub>x</sub> of a model-year 2000 (L3). In those cases, older vehicles also had an important impact on the FLMA's emission factors. In relation PM, sampled EF values over the years (0.002 to 0.001 g·km<sup>-1</sup>) have been far below PROCONVE's emission limits (0.05 in L3 to 0.025 g·km<sup>-1</sup> in L6) for light-duty vehicles, demonstrating that PM restriction has not been a target of PROCONVE for light-duty vehicles. Thus, PM emission limit for light-duty vehicles has not been encouraging fuel and automotive improvements. In contrast to the other pollutants, average EF for N<sub>2</sub>O has increased over the years. It reflects the fact that N<sub>2</sub>O is not regulated by PROCONVE.

Fig. 4 shows NO<sub>x</sub> emission factors, represented by bars for each sampling year, for heavy-duty vehicles that use diesel fuel. The average EF of the FLMA's heavy-duty vehicles is indicated by a dashed line along the entire period of PROCONVE. PROCONVE's phases are represented by text arrows showing the beginning of each phase implementation. The red curve shows the percentage of all heavy-duty vehicles in the FLMA, licensed until the corresponding year. In the figure, phases P3, P5 and P7 demonstrated the most significant reduction on NO<sub>x</sub> emission factors. As indicated by the red curve, half of the vehicles were manufactured until 2009, which has kept the average EF of heavy-duty vehicles close to phase P5. In that case, older vehicles also had a significant impact on the weighted average EF. The best case scenario would be found if the weighted EF was close to those of P7 (2012–2018).

Besides NO<sub>x</sub>, the EFs for CO, HC MP and N<sub>2</sub>O were also analyzed for heavy-duty vehicles. The average EF for CO, and HC for heavy-duty vehicles, are also equivalent to the EF of a year-model from 2000 (P3). Contrary to light-duty category, PM emission factors of heavy-duty vehicles have reduced drastically over the years. In relation to the FLMA's average PM EF, it is comparable to the EF of a year-model from 1999. N<sub>2</sub>O has not decreased, since it is not regulated by PROCONVE.

It is important to highlight that these EFs do not include deterioration factors due to vehicular use, therefore the comparison of correspondent EF was optimistic.

### 3.2. Vehicular emissions from the FLMA

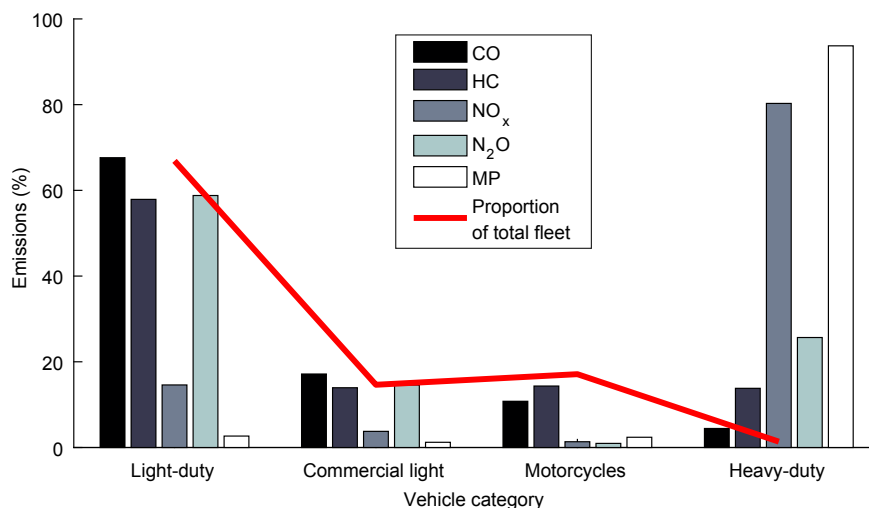
Formula (1) was applied to the weighted EFs presented in Table 2, along with the number of vehicles and distance traveled obtained from the TransCAD model to calculate categorized and total emissions from the FLMA. Emissions from the FLMA are given in Table 3, according to pollutant and vehicle category. Estimations were conducted for the peak hour in the afternoon. It can be seen that emissions are very sensitive to vehicle category, as EF can vary significantly among categories. The emissions were estimated for CO, HC, NMHC, CH<sub>4</sub>, RCHO, NO<sub>x</sub>, N<sub>2</sub>O and PM.

Fig. 5 presents the contribution of each category to the total emission by pollutant. From the figure, it is visible that light-duty vehicles emit most part of CO, HC and N<sub>2</sub>O, while heavy-duty vehicles are the biggest NO<sub>x</sub> and PM emitters. Even in lower proportion



**Table 3**  
Vehicular emissions (g km<sup>-1</sup>) from the FLMA.

| Category         | CO     | HC    | NMHC  | CH <sub>4</sub> | RCHO | NO <sub>x</sub> | N <sub>2</sub> O | PM   |
|------------------|--------|-------|-------|-----------------|------|-----------------|------------------|------|
| Light-duty       | 1271.3 | 127.3 | 105.8 | 21.5            | 3.3  | 89.9            | 8.2              | 0.5  |
| Commercial light | 322.4  | 30.7  | 25.4  | 5.3             | 0.8  | 23.2            | 2.0              | 0.2  |
| Motorcycles      | 202.8  | 31.5  | 26.4  | 5.1             | –    | 8.3             | 0.1              | 0.5  |
| Heavy-duty       | 83.4   | 30.4  | 23.2  | 7.1             | –    | 494.3           | 3.6              | 18.3 |
| Total            | 1879.9 | 219.8 | 180.8 | 39.0            | 4.1  | 615.6           | 13.9             | 19.5 |



**Fig. 5.** Percentage of category contribution to the total emissions of CO, HC, NO<sub>x</sub>, N<sub>2</sub>O and PM in the FLMA (left axis) and proportion of light-duty, commercial light, motorcycle and heavy-duty categories in relation to the total fleet of FLMA (right axis).

(less than 2% of the fleet) trucks account for a significant amount of total emissions. Commercial light category has an emission profile similar to light-duty vehicles, but with smaller magnitude, since it accounts for less than 15% of the fleet. Compared to motorcycles, commercial light category emits more CO, NO<sub>x</sub> and N<sub>2</sub>O; although motorcycles' fleet is slightly bigger. The motorcycles PM EF has only started to be informed by CETESB in 2017. As shown in Table 3 and 5, PM emissions from motorcycles are as significant as those from light-duty vehicles.

Table 4 compares emissions found in the present study (FLMA) with the emissions in The Brazilian Metropolitan Areas (MA) of Fortaleza (Policarpo et al., 2018) and Campinas (Ueda and Tomaz, 2011), the MA of Attica in Greece (Fameli and Assimakopoulos, 2015), the states of Rio de Janeiro (Souza et al., 2013) and São Paulo (CETESB, 2018) in Brazil, the North of China (Tang et al., 2016), and with the Brazilian National Inventory (Brazil, 2013). In order to determine annual emissions from the FLMA, peak hour represented 7.4% of the total flow of vehicles in a day. This simplification was based on studies conducted in the city of São Paulo (Lents et al., 2004). Seasonal variations were not accounted for.

Among all regions compared in Table 4, FLMA is the lowest emitter. It can be partially explained by the FLMA smaller fleet; however differences in method also impact on emissions estimation.

In Brazil (2013), instead of using vehicular activity modeled data, the authors estimated the circulating fleet by applying a scrap function to the licensed fleet, in order to simulate the percentage of vehicles that cease to circulate. For Otto cycle models, the scrap curves resulted from a renormalized logistic function (Brazil, 2013). The same functions were applied to the MA-Fortaleza (Policarpo

**Table 4**  
Vehicular emissions (in t·year<sup>-1</sup>) in Brazil and the world. Area (km<sup>2</sup>) and considered fleet (10<sup>3</sup>un) are also shown to allow for comparison.

| Study area              | Area (km <sup>2</sup> ) | Year | Fleet (10 <sup>3</sup> un) | CO        | NO <sub>x</sub> | PM      |
|-------------------------|-------------------------|------|----------------------------|-----------|-----------------|---------|
| FLMA                    | 1706                    | 2017 | 645                        | 9272      | 3037            | 97      |
| MA - Fortaleza          | 2575                    | 2015 | 1370                       | 72,863    | 15,036          | 604     |
| MA- Campinas            | 3792                    | 2011 | 1337                       | 244,450   | 46,350          | 2260    |
| Attica (Greece)         | 3808                    | 2010 | 3717                       | 96,108    | 27,772          | 1443    |
| State of Rio de Janeiro | 43,696                  | 2010 | 2979                       | 87,396    | 48,613          | 852     |
| State of São Paulo      | 248,209                 | 2017 | 15,235                     | 328,099   | 155,404         | 4220    |
| North of China          | 540,368                 | 2016 | 22,065                     | 9,873,454 | 1,606,470       | 115,481 |
| Brazil                  | 8,516,000               | 2013 | 41,055                     | 1,251,596 | 1,100,801       | 37,231  |

**Table 5**Emissions of CO and NO<sub>x</sub> per vehicle (kg•veh<sup>-1</sup>•year<sup>-1</sup>) and per area (t•km<sup>-2</sup>•year<sup>-1</sup>).

| Study area              | CO    | NO <sub>x</sub> | CO   | NO <sub>x</sub> |
|-------------------------|-------|-----------------|------|-----------------|
| FLMA                    | 13.9  | 4.6             | 5.4  | 1.8             |
| MA - Fortaleza          | 53.2  | 11.0            | 28.3 | 5.8             |
| MA- Campinas            | 182.8 | 34.7            | 64.5 | 12.2            |
| Áttica (Greece)         | 25.9  | 7.5             | 25.2 | 7.3             |
| State of Rio de Janeiro | 29.3  | 16.3            | 2.0  | 1.1             |
| State of São Paulo      | 21.5  | 10.2            | 1.3  | 0.6             |
| North of China          | 447.5 | 72.8            | 18.3 | 3.0             |
| Brazil                  | 30.5  | 26.8            | 0.1  | 0.1             |

et al., 2018) and the States of Rio de Janeiro (Souza et al., 2013) and São Paulo (CETESB, 2018). Scrap curves were not applied to MA-Campinas, therefore the vehicular fleet was considered to be the licensed fleet (Ueda and Tomaz, 2011). In the North of China, official statistics yearbooks provided data on vehicles per category. Classification of vehicles by fuel type was estimated based on sales reports (Tang et al., 2016). In comparison, inventories that calculate emissions directly from sales history information or licensing data without scrap curves may result in overestimation. Although the present study did not apply scrap curves to its licensing data, this data was only used to obtain fleet categorization and fuel type. This study considered modeled count of private and heavy-duty vehicles from *TransCAD* as the circulating fleet.

In relation to the annual distance traveled by each category of vehicle, Brazil (2013) highlighted that this information is scarce in Brazil as the very few studies about this subject are local studies. Therefore, in Brazil (2013) the data collected on intensity of use was adjusted according to fuel consumption (Brazil, 2013). The State of São Paulo has curves of observed intensity of use, according to vehicular categories, adapted yearly to fuel consumption (CETESB, 2018). In Rio de Janeiro, the intensity of use was based in Brazil (2011), adapted by fuel consumption in Rio de Janeiro (Souza et al., 2013). For MA-Campinas, the authors arbitrated a medium annual distance traveled, with constant yearly reduction rate (Ueda and Tomaz, 2011). For MA-Fortaleza, the same intensity of use curves designed by CETESB (2013) were applied, adjusted to fuel consumption in Fortaleza (CETESB, 2013; Policarpo et al., 2018). In the North of China, annual traveled mileages were calculated according to the road network capacity - traffic volume, volume share, actual load rate, and average load capacity (Tang et al., 2016).

About the estimates of flexible-fuel vehicles that opt to use ethanol, Brazil and the State of São Paulo used the relation of price and fuel consumption designed by Goldemberg (2008) (Brazil, 2013; CETESB, 2019). In the State of Rio de Janeiro, Souza et al. (2013) adopted a statistically adjusted function developed by Rosa (2011), which take into consideration consumers' behavior in addition to fuel market prices (Rosa, 2011). According to Rosa (2011), the function results in segregation factors similar to those of Goldemberg's curves (Goldemberg et al., 2008; Rosa, 2011; de Souza et al., 2013). MA-Campinas and MA-Fortaleza did not present their considerations concerning segregation factors for flexible-fuel technology.

Instead of EF sampled for new vehicles, Brazil (2011) developed increments to add to the EF, considering fleet deterioration per accumulated kilometers traveled. The increments need to be added to the EF in every 80.000 km (Brazil, 2011; Brazil, 2013). As in the Brazilian National Inventory, in MA-Fortaleza, the States of Rio de Janeiro and State of São Paulo applied a "deteriorated EF" for cars and light commercials (CETESB, 2018; Policarpo et al., 2018; Souza et al., 2013). Deterioration factors calculated by Ueda and Tomaz (2011) depended on accumulated kilometers traveled, estimated using an adjusted function designed for the State of São Paulo in 2004 (Ueda and Tomaz, 2011). For the North of China, the authors stated that there was no complete emission factor database for China, therefore EFs were obtained through literature research (Tang et al., 2016).

Fameli and Assimakopoulos (2015) applied the software COPERT IV to estimate vehicular emissions in Attica-Greece. Estimations were based on Tier 3 of the EMEP/EEA (2013) method. The study had detailed information on the circulating fleet of 2006. The increment from 2006 to 2010 was estimated through sales reports. About intensity of use, Fameli and Assimakopoulos (2015) applied intensity of use curves from literature in which older vehicles travel fewer kilometers (Fameli and Assimakopoulos, 2015).

According to Brazil (2011), information on the effect of real conditions of vehicular use on emissions in Brazil are scarce, as well as the intensity of use and accumulated kilometers traveled. Also, intensity of use was found to be correlated to the age of vehicles, and this information is highly dependent on social and economical indicators (Brazil, 2011). Therefore, adding deterioration factors based on intensity of use of other regions in Brazil represent a source of uncertainty that needs to be addressed in further studies. Deteriorated EFs are source of uncertainties in other countries as well, as in Attica - Greece and the North of China (Fameli and Assimakopoulos, 2015; Tang et al., 2016).

Not applying deterioration factors may have resulted in reduced emissions in the FLMA, compared to the other cities presented in Table 4. Therefore, the vehicular emissions presented for the FLMA represents the most optimistic values of emissions in the FLMA at the present, given its current fleet.

To compare emissions in Table 4 with different order of magnitude, a ratio between annual emissions and fleet, and between annual emissions and territory were calculated (Table 5).

The North of China, the greatest emitter of all pollutants in Table 4, also presented the biggest ratios of emission per vehicle. Besides the huge fleet, vehicles in the North of China may have high average EF and/or increased intensity of use. The MA-Campinas presented high emissions of CO per vehicle, being five times superior to Brazil. Increased emissions in MA-Campinas may be related to not using scrap curves and/or to the value arbitrated as accumulated distance travelled. The states of Rio de Janeiro and São Paulo,

MA-Fortaleza, Attica - Greece and the FLMA presented emission per vehicle close to Brazil for CO, but lower for NO<sub>x</sub>.

In relation to the ratio of emissions per area, Brazil and the states of Rio de Janeiro and São Paulo presented the lowest values found for both CO and NO<sub>x</sub>. The FLMA had the 4th smallest ratios for both NO<sub>x</sub> and CO. In general, the magnitude order of NO<sub>x</sub> emissions per area had little variation amongst regions. For CO, amplitude of values was more significant. Ratios from MA-Fortaleza, Attica - Greece and the North of China were closer, and MA-Campinas was the greatest in this case.

In general, NO<sub>x</sub> emission ratios had less variation than CO. Distinctions in results may be explained by differences in the applied methods among studies, fleet composition and age and spatial resolution.

### 3.3. Spatial distribution

The geographical distribution of emissions from the FLMA, as described in Section 2.5, is illustrated by Fig. 6. Emissions are split in light category (light-duty, commercial light and motorcycles) on top and heavy category at the bottom. Fig. 6 allows one to detect where emissions are critical and which category is the biggest emitter depending on the pollutant. The reader can identify the types of roads in Fig. 1.

An analysis of emissions distribution throughout FLMA's road network is shown in Fig. 7. In the figure, the variability of emissions per road type is represented by box plots, in which the inner marks in the boxes represent the medians, and the top and bottom edges of the boxes indicate the 75th and the 25th percentiles, respectively. Outliers are indicated in red, and boxes are plotted for each pollutant and according to network hierarchy. Fig. 7 also informs emissions per hierarchy and their contribution percentage on total emissions.

The traffic of vehicles according to road hierarchy was also analyzed, presented in Fig. 8. In this case, the box plots show the variability of the amount of vehicles per hour traveling on each road.

Through Fig. 6, the influence of vehicle category on pollutants emitted is noticeable. The highest emissions (more evident for NO<sub>x</sub> and MP) occur in expressways, where heavy-duty vehicles account for 22% of the traffic, as indicated in Fig. 8. It can be confirmed by Fig. 7, where expressway hierarchy was responsible for 79% of NO<sub>x</sub> and over 85% of MP emissions. For other pollutants, expressways contribution was close to 50%, even though they cover fewer roads. It must be emphasized that only in 1994 emission rates started to be restricted in Brazil for heavy-duty vehicles. In this context, there is some room for improvement.

Arterial roads also presented high emissions, as can be noticed in Fig. 6, resulting in important amounts of CO, HC and N<sub>2</sub>O emitted. In arterials, the light category composed 96% of vehicular traffic. The intense flow of vehicles on arterial roads can be pointed as the main reason for their relevance. Emissions on collector roads were not as significant, even though they comprehended a significant amount of roads. With 97% of its traffic composed by light category, emissions on collector roads were not as high as arterial hierarchy as its flow of vehicles was not intense. Emissions on connector roads were not significantly smaller than on collector, despite of the difference in number of roads. Although the amount of urban roads was the absolute highest, total emissions in this type of road were not as high. Most of urban ways presented null emissions, when the red marks on Fig. 7 are considered outliers.

Residential regions with high population density presented more pollutant roads than expressways and arterials surroundings. In general, roads with intense flow of vehicles emitted more pollutants. It does not indicate, however, a region with a large amount of low emission roads (secondary roads) do not cause impact

### 3.4. Comparing current and hypothetical scenarios

A hypothetical scenario was proposed to evaluate potential reduction of emissions, in comparison to emissions in Table 3. All vehicles in the hypothetical scenario were considered to be manufactured in 2017. Estimated values and their reduction percentage are presented in Table 6.

By applying emission factors from 2017 and the current number of vehicles, emissions would decrease drastically for most of pollutants. The most significant decrease was found for NMHC results, which would decrease by 93%. This practice is already discussed for some real cases. Andrade et al. (2017) mention that would be an obvious step to scrap old vehicles from 10 to 15 years of age in most megacities (Andrade et al., 2017).

N<sub>2</sub>O presented a slight increase in emissions. This projection reflects the EF reduction discrepancies among regulated pollutants (CO, HC, NO<sub>x</sub> and MP) and N<sub>2</sub>O, which is not regulated.

In the United States of America, policies for reduction of NO<sub>x</sub> emissions had important impact on air quality control. On the other hand, the unrestricted pollutant ammonia was found to be increasing. Indeed, NO<sub>x</sub> is reduced to ammonia through vehicles catalytic converters (Li et al., 2016). Therefore the effectiveness of public policies towards emissions reduction should also be assessed for unrestricted pollutants.

Fig. 9 illustrates both current and hypothetical scenarios for comparison. Differently from Fig. 6, emissions from light and heavy categories were summed up, therefore current scenario represents total emissions for each air contaminant in the FLMA. Similarly, the hypothetical scenario shows total emissions when EFs from 2017 are applied.

Fig. 9 shows a significant reduction of emissions on roads. The hypothetical scenario showed that fleet renovation would have a huge impact on emissions, especially over major roads. However this projection is not conclusive in relation to impacts on ambient air quality caused by vehicular emissions.

In Fig. 10, emissions from current and hypothetical scenarios were distributed in pixels of 1 km<sup>2</sup>. National and state highways concentrated most of emissions (red paths in current scenario). Fig. 10 also shows percentage of reduction from up to 95% in many

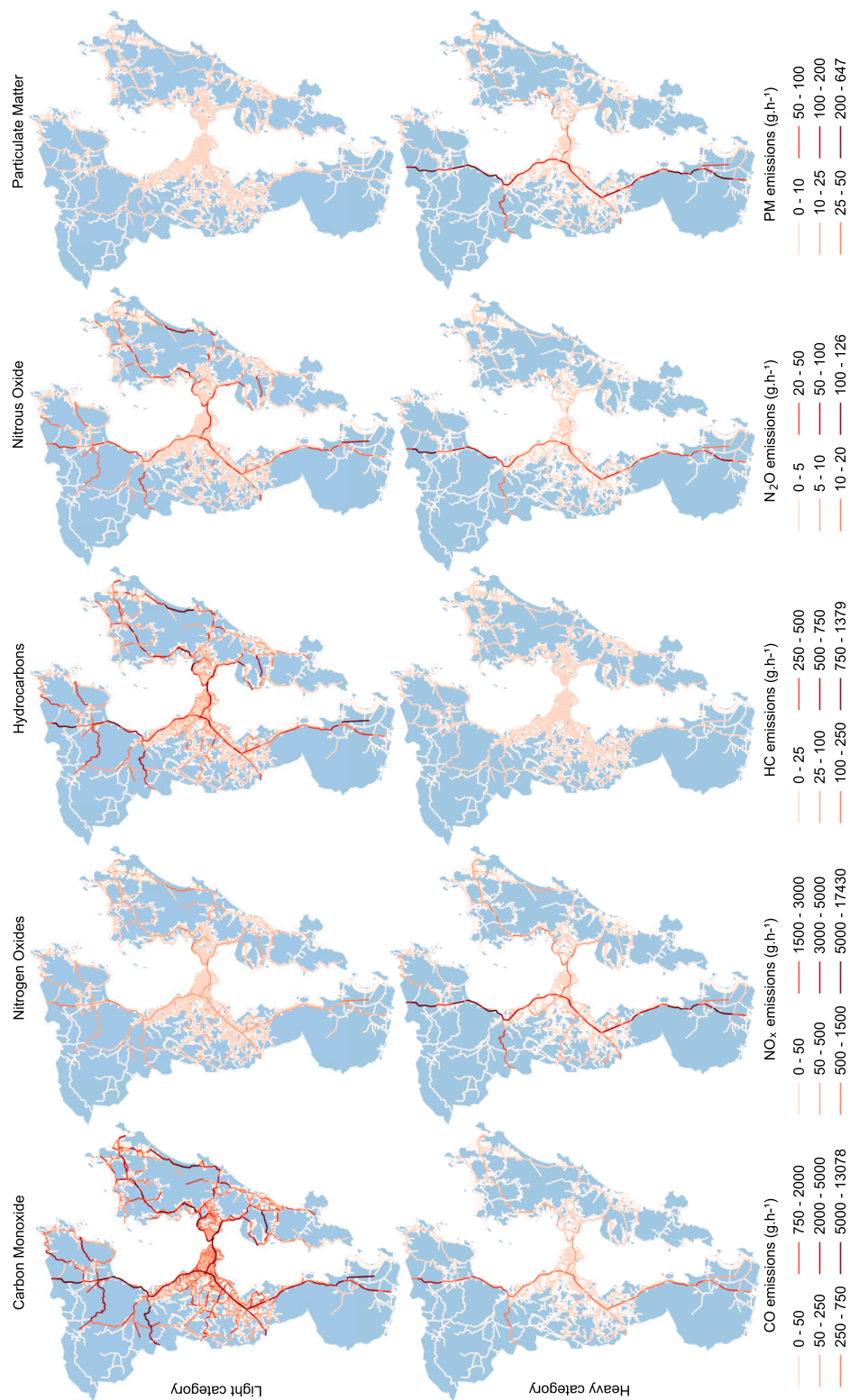
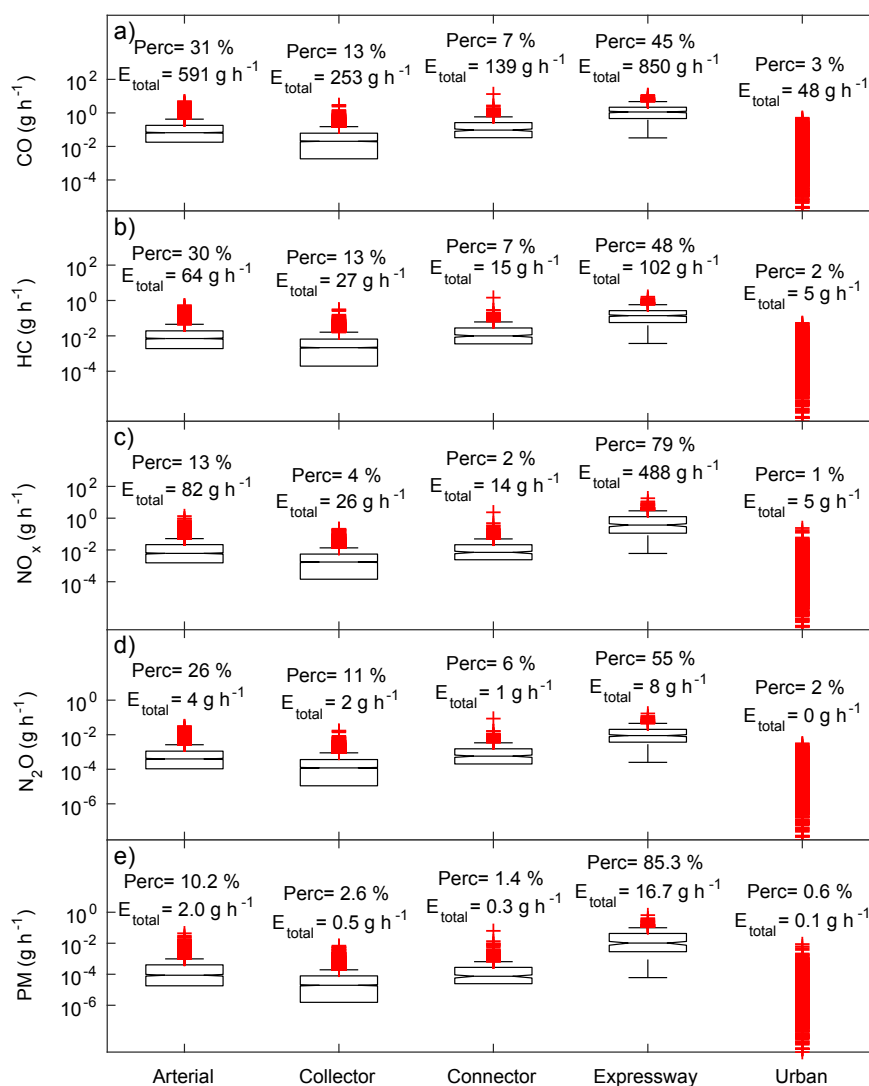


Fig. 6. Spatial distribution of CO,  $\text{NO}_x$ , HC,  $\text{N}_2\text{O}$  and MP emissions (in  $\text{g h}^{-1}$ ) from Light and Heavy categories in FLMA.



**Fig. 7.** Emissions of (a) CO, (b) HC, (c) NO<sub>x</sub>, (d) N<sub>2</sub>O and (e) PM according to network hierarchy. The boxes represent the variability of emissions per road. The inner marks in the boxes are the medians, and the top and bottom edges of the boxes indicate the 75th and the 25th percentiles, respectively. Outliers are plotted in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

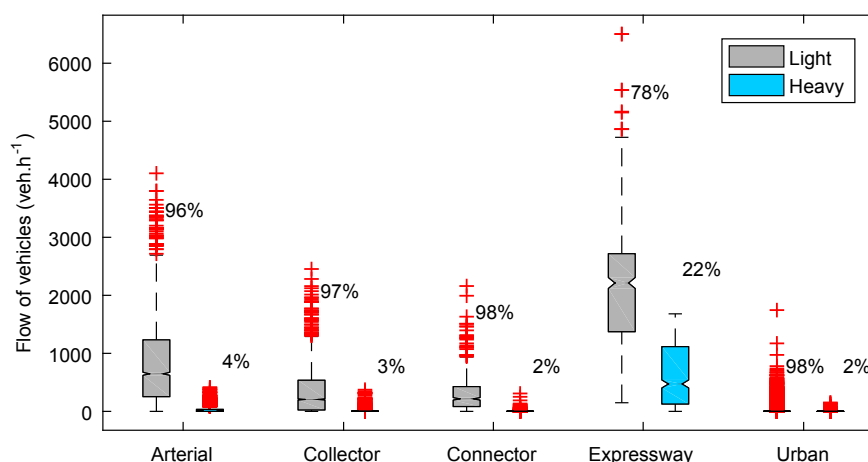
locations within the FLMA. Even on hypothetical scenario, emissions from 1 to 5 kg/h were found in places marked by dense transportation network and/or high flow of vehicles. So even with better automobile technologies (reflected by reduced EF), air quality control is still complex in places of dense network and intensive traffic.

#### 4. Conclusions

This paper proposes a method to develop high resolution (temporal and spatial) vehicular emissions inventories. The method is sensitive to the category, model-year, fuel type and flow of vehicles. Throughout the proposed technique, it is possible to produce a detailed emissions inventory for metropolitan areas and elaborate hypothetical scenarios to support decision making.

The method was tested in the Florianópolis Metropolitan Area. Licensed vehicles were classified in: light-duty, commercial light, motorcycles and heavy-duty. It allowed a characterization of emissions according to fleet composition. As the weighted emission factors were significantly increased by old vehicles, a hypothetical scenario was employed to detect potential reductions related to fleet renovation.

Having access to modeled data on vehicular flow was essential to apply this method and to be able to produce an inventory in high resolution. In this case, a detailed transport model in TransCAD has shown to be a resourceful tool for environmental purposes. Providing the necessary data on road-link resolution, aggregated to a geographically referenced vector file, road transport models are



**Fig. 8.** Flow (veh/h) categorized in light and heavy. The boxes represent the variability per type of road. The inner marks on the boxes are the medians, and the top and bottom edges of the boxes indicate the 75th and the 25th percentiles, respectively. Outliers are plotted in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

**Table 6**

Total emissions on current and hypothetical scenarios ( $\text{kg}\cdot\text{h}^{-1}$ ) and reduction (%).

| Scenario           | CO     | HC    | NMHC  | CH <sub>4</sub> | RCHO | NO <sub>x</sub> | N <sub>2</sub> O | PM   |
|--------------------|--------|-------|-------|-----------------|------|-----------------|------------------|------|
| Current            | 1879.9 | 219.8 | 180.8 | 39.0            | 4.1  | 615.6           | 13.9             | 19.5 |
| Hypothetical       | 145.3  | 23.7  | 13.2  | 10.5            | 0.6  | 163.4           | 14.1             | 2.0  |
| Emission reduction | 92.3   | 89.2  | 92.7  | 73.0            | 85.7 | 73.5            | −1.3             | 89.6 |

fit to be applied in air quality studies. It is also particularly interesting for areas that lack of vehicle activity data from monitoring systems.

Through a bottom-up approach, it was estimated that the FLMA fleet emitted approximately 1879.9 kg of CO, 180.8 kg of NMHC, 39.0 kg of CH<sub>4</sub>, 4.1 kg of RCHO, 615.6 kg of NO<sub>x</sub>, 13.9 kg of N<sub>2</sub>O and 19.5 kg of MP during its peak hour. Heavy category was responsible for most of NO<sub>x</sub> (80%) and almost all MP (94%) emitted. In relation to light category, it contributed to 96% of CO, 87% of NMHC, 82% of CH<sub>4</sub> and 74% of N<sub>2</sub>O. Vehicular source apportionment was consistent to inventories elaborated for Brazil (Brazil, 2013) and for the State of São Paulo (CETESB, 2018).

The method allowed to verify that emissions were sensitive to fleet composition, in terms of model-year and vehicular category. Even though in smaller number, old vehicles had a significant impact on pollutant emissions, as recently manufactured vehicles presented EF values up to twenty times smaller than old vehicles. This indicates that shifting old vehicles for new ones would be in line with the established practices in most metropolitan areas. Emissions distribution also responded to network hierarchy, showing that expressways, main route for heavy-duty vehicles and characterized by intense traffic, comprehended 45% of CO, 48% of HC, 79% of NO<sub>x</sub>, 55% of N<sub>2</sub>O and 85% of PM total emissions. On the other hand, urban roads presented the lowest total emissions, even though they were in greater number. Therefore, the present study was able to locate high emission roads, as well as high density of roads with low emissions.

A hypothetical scenario applying 2017 emission factors for the FLMA's vehicles would decrease emissions up to 95%, depending on pollutant. This suggests that reducing average fleet age is a possible mitigation measure. Even reducing vehicles' EF, expressways concentrated most emissions. So decreasing the amount of vehicles in circulation should also be assessed. In particular, results from expressways showed that focusing on emissions control strategies for heavy-duty vehicles could result in a significant decrease of total emissions in the FLMA of NO<sub>x</sub> and PM.

The present study also provided new information about the FLMA's vehicular emissions, which can be applied in public management. Implementation of air quality monitoring stations, cycle routes, traffic restriction of vehicles in strategic places and control of high-emitter vehicles are examples of applications that can be found in literature (Borge et al., 2014; Gulia et al., 2018; Minet et al., 2018; Popoola et al., 2018; Yang and Liu, 2018). This research also provided input data for dispersion models, allowing one to assess the impact of vehicular emissions and guiding air quality management measures.

Monitoring of emissions and air quality is almost nonexistent in Brazil, therefore estimation methods are a viable way to obtain information. However, availability of data is a limiting factor.

About the limitations of the present study, it is known that not all licensed vehicles are part of circulating fleet, yet observational data on daily traffic were not available to estimate a circulating fleet. Also, vehicle speed, intensity of use, wear in brakes and tyres and soil dust resuspension were not considered, although they would increase emissions estimation.



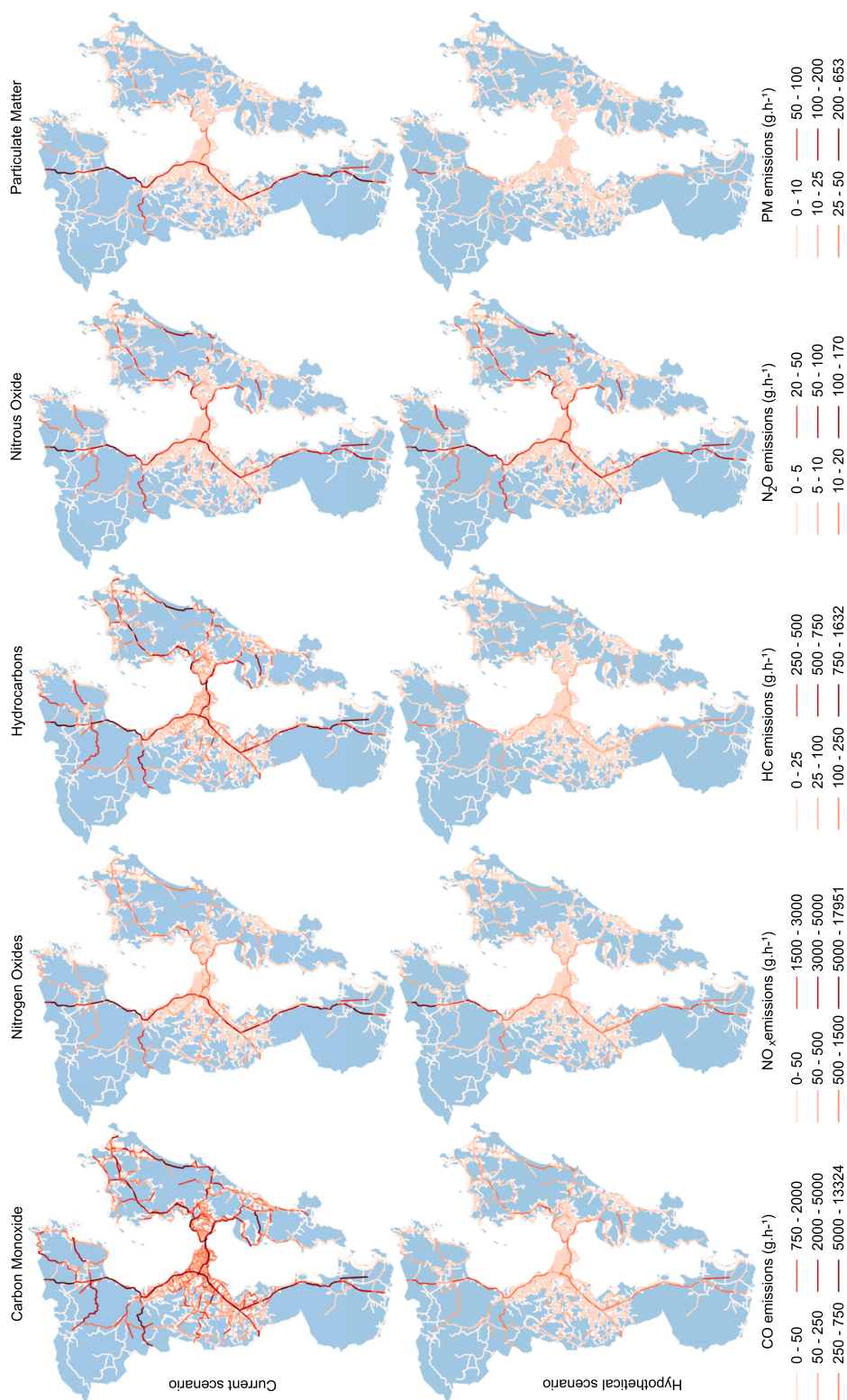


Fig. 9. Spatial distribution of CO, NO<sub>x</sub>, HC, N<sub>2</sub>O and MP total emissions (in  $\text{g}\cdot\text{h}^{-1}$ ) from Current and Hypothetical scenarios in the FLMA.

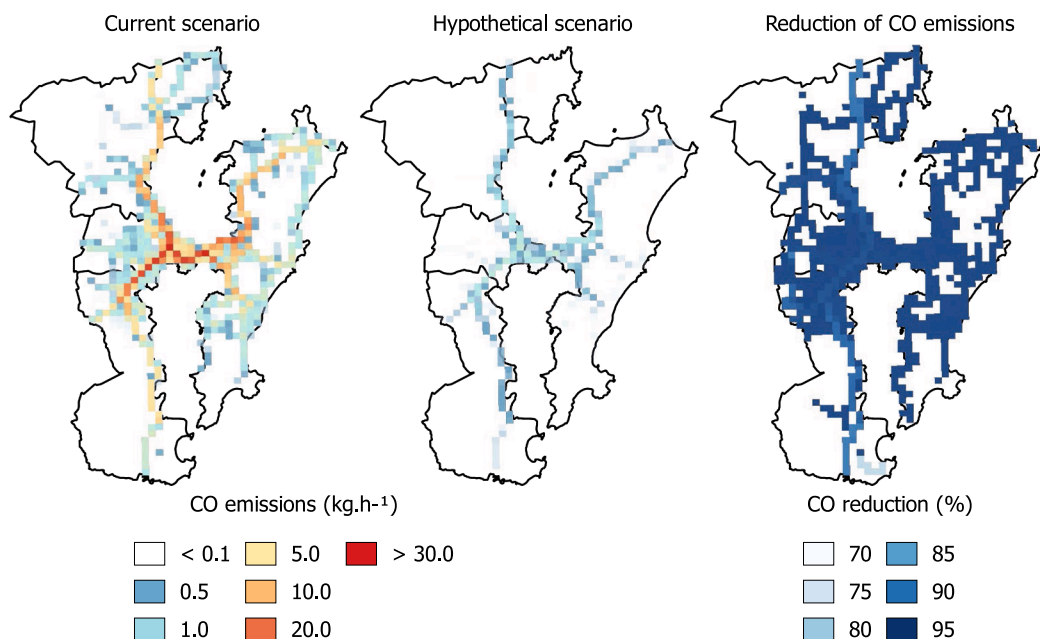


Fig. 10. Comparison between current and hypothetical CO emissions in the FLMA.

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## Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.trd.2019.10.007>.

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